# MagicCode Super-Resolution Software Migration Guide

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## 1. Purpose

This guide helps customers migrate to MagicCode Super-Resolution Software v3.0.0, including from v2.2.x / v2.1.0 / v2.0.0 and from v1.2.x image-only public docs. The v2.x → v3.0.0 ABI break is the required section for current ship.

## 2. Migrating from v2.x to v3.0.0

| Area | v1.2.x / v2.0.x / v2.1.0 | v2.2.0 | Action |
|---|---|---|---|
| Version | v1.2.0 / v2.0.0 / v2.1.0 | v2.2.0 | Rebuild against new headers/libs. |
| Modes | SPEED_MODE / BALANCED_MODE (image) | SPATIAL_* plus TEMPORAL_SPEED_MODE and TEMPORAL_BALANCED_MODE | Rename image enumerators; add 2/3 only for graphic. |
| MC_Process | Some docs showed (handle, in, out) | mc_nscaler_enable(&handle, &in_frame, &out_frame) | Split in/out frames; graphic needs frame. |
| Scale | [1,8] for documented image | Image [1,8]; graphic (1,8] | Reject graphic 1.0. |
| Windows | OpenGL often documented alone | Vulkan / D3D11 / OpenGL 4.3 graphic on session API | Enable default remains OpenGL; D3D11 is session-only. |
| Enable helper | v2.1.0 MC_Enable image helper | Removed in v2.2.0 | Default-create via mc_nscaler_enable (NULL handle). |
| Public API | MC_Init / MC_Process / MC_Uninit / MC_Control | mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable / mc_nscaler_version | Rename calls; create via SET_PARAM or NULL-handle enable. |
| Create params | input_param_t (struct_size, width, height) | ctrl_param_t; size on mc_nscaler_input_frame_t | Drop struct_size; move width/height to the frame. |
| Frame struct | magic_frame_t (image_in/image_out) | mc_nscaler_input_frame_t + mc_nscaler_output_frame_t | Split input/output; temporal_frame_t stays on .frame. |
| Platforms | Android, iOS/iPadOS/macOS, Windows | + Linux x86_64 (CPU image archive) | Link lib/linux/libmagic_sr.a. |
| mc_nscaler_enable (v3) | mc_nscaler_enable(&h, &in, &out) | mc_nscaler_enable(&h, in2, &out) with in2[2] | Pass two pointers; in2[1]=NULL unless GEN interpolate. |
| alg_mode_e (v3) | SPATIAL_* / | SR_*_FOR_IMAGE/GRAPHIC | Rename enumerators; |

| | | | |
|---|---|---|---|
| | TEMPORAL_* ; MAX_ALG_MODE=4 | plus GEN bits; MAX_ALG_MODE=16 | values 0..3 unchanged. |
| Frame generation | Not in v2.2 public nscaler | GEN_SPEED / GEN_BALANCED on Vulkan + MAGIC_SR_HAS_FG | Ship magic_fg_core.bin for CoreV3; Android Vulkan has no FG. |
| Version / unload | Static string v2.2.0 | Static string v3.0.0; invalid after FreeLibrary/dlclose | Copy before unload. DLL exports six symbols. |

Required changes from v2.2.x (and any v2.x nscaler caller) to v3.0.0:

## Enumerator mapping (v2.x → v3.0.0)

- SPATIAL_SPEED_MODE (0) → SR_SPEED_MODE_FOR_IMAGE (0)
- SPATIAL_BALANCED_MODE (1) → SR_BALANCED_MODE_FOR_IMAGE (1)
- TEMPORAL_SPEED_MODE (2) → SR_SPEED_MODE_FOR_GRAPHIC (2)
- TEMPORAL_BALANCED_MODE (3) → SR_BALANCED_MODE_FOR_GRAPHIC (3)
- New: GEN_SPEED_MODE_FOR_GRAPHIC (4), GEN_BALANCED_MODE_FOR_GRAPHIC (8). MAX_ALG_MODE 4 → 16. Combination example: SR_SPEED_MODE_FOR_GRAPHIC|GEN_SPEED_MODE_FOR_GRAPHIC = 6.

## mc_nscaler_enable signature

v2.x:

```
int mc_nscaler_enable(void **handle, mc_nscaler_input_frame_t *in_frame,
mc_nscaler_output_frame_t *out_frame);

mc_nscaler_enable(&handle, &in_frame, &out_frame);
```

v3.0.0:

```
int mc_nscaler_enable(void **handle, mc_nscaler_input_frame_t *in_frame[2],
mc_nscaler_output_frame_t *out_frame);

mc_nscaler_input_frame_t *in2[2] = { &in_frame, NULL };

mc_nscaler_enable(&handle, in2, &out_frame);
```

- Passing &in_frame as the second argument is no longer valid. For GEN interpolate, set in2[1] to the post-interpolate frame; for SR leave it NULL.

## Common errors after upgrade

- Compile error on mc_nscaler_enable: update the two-frame pointer array.
- Unresolved SPATIAL_* / TEMPORAL_* identifiers: apply the enumerator mapping.
- -100050 / -102013: GEN requested on a non-Vulkan backend or a build without MAGIC_SR_HAS_FG (including current Android Vulkan).

- -100051 / -102014: GEN_BALANCED weights missing. Set ctrl_param_t.model_path to the FG weight file, or leave it empty and place model/magic_fg_core.bin / set MAGIC_FG_WEIGHT_PATH.
- Unexpected handle recreate on GEN: in_frame[0].width/height must be the render input size, not the output color size.
- Crash after FreeLibrary/dlclose while reading mc_nscaler_version(): copy the string first.
- Vulkan Speed (Lite-TAAU) now lets an explicit strong reactive mask suppress the still-camera accumulation branch. Previously that branch ignored the caller reactive mask when the camera was still or per-pixel MV was below about 0.2 px, so pixels with ~0 MV but changing content (TV / video-wall / LED-screen textures) could keep ghosting for about 64–96 frames after a content cut. Strong reactive (≥ about 0.85, fully effective at ≥ 0.95) now weakens that accumulation. Omitting reactive or passing a weak value (< 0.85) is unchanged. Recommendation: mark screen-like objects whose MV is 0 but whose content changes (for example with CustomDepth stencil) as reactive at strength 1.0. Limit: Vulkan only in this release. The Metal / GLES / D3D11 Speed still-accumulation structure is the same idea but is not modified or verified here.

## Graphic Super-Resolution

Graphic super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x SR_SPEED_MODE_FOR_IMAGE) and processes the first frame. Pass mc_nscaler_input_frame_t *in2[2] = { &in, NULL } (set in2[1] only for GEN interpolate).

### Modes

| Value | Enumerator | Role |
|---|---|---|
| 0 | SR_SPEED_MODE_FOR_IMAGE | Image, throughput first |
| 1 | SR_BALANCED_MODE_FOR_IMAGE | Image, quality/cost trade-off |
| 2 | SR_SPEED_MODE_FOR_GRAPHIC | Graphic, throughput first |
| 3 | SR_BALANCED_MODE_FOR_GRAPHIC | Graphic, canonical FSR-family path |
| 4 | GEN_SPEED_MODE_FOR_GRAPHIC | FG-Fast (analytic, no weights). Vulkan + MAGIC_SR_HAS_FG. Combinable with SR bits; bare 4 equals SR_SPEED_MODE_FOR_IMAGE\|GEN_SPEED. |
| 8 | GEN_BALANCED_MODE_FOR_GRAPHIC | FG-CoreV3. Same availability as Fast. Needs magic_fg_core.bin. Do not set both GEN bits. |

- Graphic scaler_factor is (1, 8]; 1.0 itself is rejected. Image remains [1.0, 8.0].
- SR_SPEED_MODE_FOR_GRAPHIC on Metal, Vulkan, and OpenGLES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family graphic path (same family as Balanced, not a separate customer API).

- SR_BALANCED_MODE_FOR_GRAPHIC uses the canonical FSR-family path on every GPU graphic backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLES do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run graphic modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

## Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at create (mc_nscaler_control SET_PARAM, or the first mc_nscaler_enable): Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan graphic requires mc_nscaler_input_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on ctrl_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

## Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both graphic modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

## 3. Native code migration

1. Replace headers with v3.0.0 mc_interface.h (mc_enable.h is removed). Public MC_ERROR_* codes are in mc_error.h, included by mc_interface.h.
2. Rebuild against v3.0.0 libraries (Windows: libmagic_sr.dll / .lib / .dll.lib; Linux: libmagic_sr.a / .so with Vulkan Graphic (Vulkan Temporal)/TAAU + FG).
3. Set struct_size on temporal_frame_t (ctrl_param_t has no struct_size; the image size moves to mc_nscaler_input_frame_t).
4. Replace MC_Init / MC_Process / MC_Uninit and MC_Enable calls with mc_nscaler_control(SET_PARAM) create, mc_nscaler_enable(in2[2]) per frame, and mc_nscaler_disable at shutdown.
5. For graphic, import gpu_context and pass depth, motion, jitter, camera, and (Vulkan) a recording command_buffer.
6. Use QUERY_STATUS; inspect temporal_* fields only when temporal_status_valid is set.

## 4. Unity / Unreal

Replace plugin binaries as you republish. Plugin image APIs do not feed graphic extras. Native v3.0.0 graphic is the nscaler session API (mc_nscaler_enable with in2[2] and temporal_frame_t). FG is the GEN bits on the same API.

## 5. Documentation migration

- Use User_Guide.md / 用户使用说明书.md v3.0.0 for customer integration.
- Legal documents (EULA, Privacy Policy, Data Reporting Notice) keep their existing substance and are not version-bumped by this technical drop.
- Filenames under doc/版本文档 no longer include a version suffix.

## 6. Acceptance checklist

- mc_nscaler_version returns v3.0.0 (copy the string before unloading the module).
- Image processing works on target mobile backends.
- If shipping graphic: SR_SPEED_MODE_FOR_GRAPHIC and SR_BALANCED_MODE_FOR_GRAPHIC process with temporal_frame_t depth/motion/jitter. If shipping FG: GEN bits, two-frame enable, Vulkan, and weights for GEN_BALANCED.
- CPU graphic is rejected as expected.
- Linux x86_64 archive links CPU image and Vulkan Graphic (Vulkan Temporal)/TAAU + FG.
- Known Issues reviewed.